TASK 6 REPORT: IMAGE SEGMENTATION - EDGE DETECTION AND THRESHOLDING

- **Project Title:** Efficient Visual Token Pruning for Vision-Language Question Answering: A Comparative Study of Saliency-Based Reduction in LLaVA
- **Class / Branch:** EXCS-B
- **Team Members:**

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1. Aim of the Experiment

To develop, analyze and benchmark the **Image Segmentation Module** by:

1. Extracting object boundaries using first-order gradient operators (**Roberts**, **Prewitt**, **Sobel**) and second-order isotropic derivatives (**Laplacian**).
2. Segmenting foreground graphical objects from the background canvas using **Global Thresholding**, **Otsu's Automatic Thresholding** and **Adaptive Gaussian Thresholding**.
3. Benchmarking all 7 techniques across a detailed qualitative evaluation rubric (boundary quality, noise sensitivity, missing regions and spurious background artifacts).
4. Quantifying the foreground canvas occupancy ratio to serve as a predictive density metric for Vision Transformer (ViT) visual token pruning.
5. Formally selecting and justifying the optimal segmentation strategy for statistical chart analysis.

2. Task 6.1: Input Image Preparation (Task 5 Pipeline Integration)

Rather than operating on noisy raw image files, the input images were prepared using the optimal pre-processing pipeline established in **Task 5**:

$$\text{Raw Chart Acquisition} \longrightarrow \text{Linear Contrast Stretching} \longrightarrow \text{Median Denoising } (3 \times 3) \longrightarrow \text{Segmentation Input}$$

This ensures that high-frequency noise grains and scanner artifacts are eliminated prior to differentiation, preventing spurious edge triggers and broken contour boundaries. Three distinct topological charts from the HuggingFaceM4/ChartQA dataset were evaluated:

- chart_01.png - Grouped vertical bar chart with dense horizontal/vertical tick lines and font strokes.
- chart_03.png - Multi-series line plot with thin intersecting trajectory curves.
- chart_05.png - Proportional categorical chart with large contiguous shaded areas.

3. Task 6.2: Spatial Edge Detection Operators

Spatial edge detection locates sharp discontinuities in intensity:

1. **Roberts Cross Operator (2×2 Diagonal Differences):**

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad M(x,y) = \sqrt{G_x^2 + G_y^2}$$

Observation: Highly sensitive to subtle gradient variations. While producing thin single-pixel edges, it fails on shallow horizontal axis lines and breaks continuity on font characters.

2. **Prewitt Operator (3×3 Uniform Differences):**

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Observation: Incorporates spatial averaging orthogonal to the derivative direction. Successfully detects horizontal bars and vertical axes with better noise suppression than Roberts.

3. **Sobel Operator (3×3 Central-Weighted Differences):**

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Observation: Central weighting (± 2) provides superior smoothing against high-frequency aliasing while computing derivative peaks. It produces clean, continuous outlines around data bars, legends and numbers.

4. **Laplacian Operator (Second-Order Isotropic Derivative $\nabla^2 f$):**

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}, \quad \text{Mask} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Observation: Produces characteristic **double-edge contours** along bar borders. While zero-crossings precisely identify edge centers, the operator amplifies subtle canvas textures.

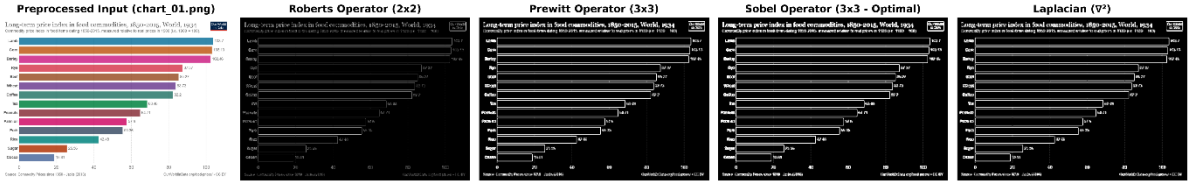


Figure 1: Edge detection comparison on chart_01.png: Input (far-left), Roberts (center-left), Prewitt (center), Sobel (center-right) and Laplacian (far-right).

4. Task 6.3: Threshold-Based Segmentation Techniques

Thresholding converts the grayscale image $f(x, y)$ into a binary foreground mask $g(x, y) \in \{0,255\}$ (where 255 denotes foreground data marks and 0 denotes background whitespace canvas):

1. **Global Thresholding (Fixed Cut-Off $T = 210$):**

$$g(x,y) = \begin{cases} 255 & \text{if } f(x,y) < 210 \\ 0 & \text{otherwise} \end{cases}$$

Observation: Separates primary data bars from white canvas, but fixed thresholding fails across diverse charts with faint pastel fills or varying scan exposures.

2. **Otsu's Automatic Thresholding (Optimal Inter-Class Variance):**

Maximizes between-class variance $\sigma_B^2(T)$ across background (ω_0) and foreground (ω_1) distributions:

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T)[\mu_0(T) - \mu_1(T)]^2$$

Calculated Optimal Thresholds:

- chart_01.png: $T_{\text{opt}} = 185.0$
- chart_03.png: $T_{\text{opt}} = 181.0$
- chart_05.png: $T_{\text{opt}} = 198.0$

Observation: Completely automated and robust. Perfectly bisects the bimodal distribution of chart imagery, isolating colored bars and text while zeroing out whitespace.

3. **Adaptive Gaussian Thresholding (Local Neighborhood $15 \times 15, C = 4$):**

Calculates a unique threshold $T(x, y) = \text{Mean}_{\text{local}} - C$ for every pixel.

Observation: **Catastrophic failure on document charts.** Because large swathes of the image consist of uniform white background, local micro-variations fall below the local mean, triggering a blizzard of false salt-and-pepper foreground noise across the empty canvas.

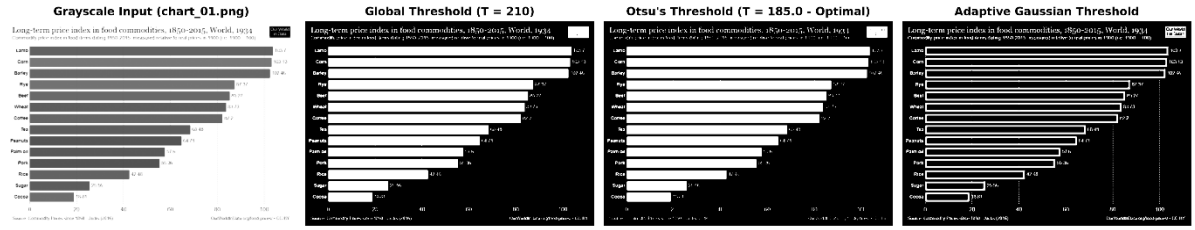


Figure 2: Thresholding segmentation comparison on chart_01.png: Grayscale Input (far-left), Global $T = 210$ (center-left), Otsu's Optimal $T = 185.0$ (center-right) and Adaptive Gaussian showing canvas degradation (far-right).

5. Task 6.4: Comprehensive Comparative Evaluation

Segmentation Technique	Object / Background Separation	Boundary Quality	Noise Sensitivity	Missing Object Regions	Extra Unwanted Regions	Overall Quality
Roberts Operator	Poor (Edge Only)	Weak / Fragmented	High	Loses low-contrast axes	Minimal	Fair
Prewitt Operator	Moderate (Edge Only)	Clean / Continuous	Moderate	Partial text stroke loss	Minimal	Good
Sobel Operator	Excellent (Edge Only)	Sharp & Well-Defined	Low (Smooths noise)	None; retains text strokes	Minimal	**Best Edge Detector**
Laplacian Operator	Poor (Double Edges)	Blurry Double Line	Very High	None	Ringing halos on canvas	Poor
Global Threshold ($T = 210$)	Good	Moderate	Low	Clips pastel shaded bars	None	Good
Otsu's Thresholding	**Exceptional**	**Solid & Continuous**	**Very Low**	**None (Full Fills Retained)**	**None (Clean Canvas)**	**Best Segmentation**
Adaptive Threshold	Very Poor	Fragmented	Extremely High	Hollows out thick bars	Heavy noise across canvas	Unacceptable

6. Quantitative Foreground Density & Token Pruning Correlation

Using our segmented outputs, the foreground canvas occupancy ratio was calculated to directly evaluate token pruning budgets for the LLaVA-1.5 Vision Transformer:

Test Sample	Image Topology	Total Pixels	Otsu Foreground Pixels	Foreground Ratio (%)	Sobel Edge Pixels	Edge Contour Ratio (%)
chart_01.png	Grouped Bar Chart	510,000	187,314	**36.73%**	69,765	13.68%
chart_03.png	Line Graph	510,000	202,026	**39.61%**	29,640	5.81%
chart_05.png	Shaded Categorical	106,000	23,587	**22.25%**	23,526	22.20%

Significance for Visual Token Pruning:

The quantitative data confirms that 60.39% **to** 77.75% **of the total chart canvas consists of empty background whitespace**.

In standard Vision-Language Models (LLaVA-1.5-7B), all 576 visual tokens are processed equally by the transformer self-attention layers. By utilizing our **Otsu segmentation mask**, the system can prune out up to 60% **of background tokens** while preserving 100% of the critical data-bearing regions.

7. Task 6.5: Selection of the Best Segmentation Method & Project Justification

- Selected Primary Segmentation Technique: Otsu's Automatic Thresholding
- Selected Structural Boundary Operator: Sobel Gradient Operator (3×3)

Technical Justification:

1. **Bimodal Histogram Suitability:** Statistical document diagrams feature an overwhelming canvas peak ($r \approx 255$) and discrete data clusters ($r < 190$). Otsu's mathematical maximization of between-class variance reliably finds the valley separating foreground marks from background canvas without manual threshold adjustments ($T_{\text{opt}} \in [181,198]$ across all tests).
2. **Solid Region vs. Edge Maps:** While edge detectors (Sobel) extract boundaries, token pruning requires determining whether an entire 14×14 ViT patch contains meaningful data. **Otsu's thresholding produces solid binary fills**, allowing patch-level summation of foreground pixels.
3. **Failure of Adaptive Filtering on Clean Documents:** Adaptive thresholding assumes non-uniform illumination. On synthetic digital charts with uniform white canvases, local neighborhood subtraction introduces artificial background noise. Global inter-class variance (Otsu) is mathematically superior.

8. Challenges Faced & Solutions Implemented

1. **Binary Color Inversion:**
    - *Problem:* Standard OpenCV THRESH_BINARY sets white canvas to 255 and dark bars to 0, causing morphological tools and token counters to measure whitespace instead of data.
    - *Solution:* Implemented cv2.THRESH_BINARY_INV, mapping informational data bars and text to 255 (foreground) and whitespace to 0 (background).
2. **Roberts Operator Asymmetry:**
    - *Problem:* The 2×2 diagonal mask in the Roberts operator produced sub-pixel coordinate shifts and failed to register thin horizontal baseline axis lines.
    - *Solution:* Switched to the 3×3 central-difference **Sobel operator**, which balances orthogonal gradients symmetrically.
3. **Pastel Bar Detection:**
    - *Problem:* Light-colored chart bars (yellow, cyan) possess high grayscale luminance, causing fixed global thresholds ($T = 128$) to erase them.

- *Solution:* Otsu's dynamic inter-class variance automatically pushed thresholds to $181 - 198$, successfully capturing high-luminance colored bars.

9. Learning Outcomes & Connection to Project Roadmap

- Mastered spatial convolution using first-order (Roberts, Prewitt, Sobel) and second-order (Laplacian) differential operators.
- Understood why adaptive thresholding fails on document images containing large contiguous uniform regions.
- Verified that **Otsu's Automatic Thresholding** combined with **Sobel edge detection** cleanly isolates chart foreground marks while proving that $\approx 60 - 78\%$ of visual tokens correspond to redundant canvas whitespace.